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Build a Virtual Private Cloud

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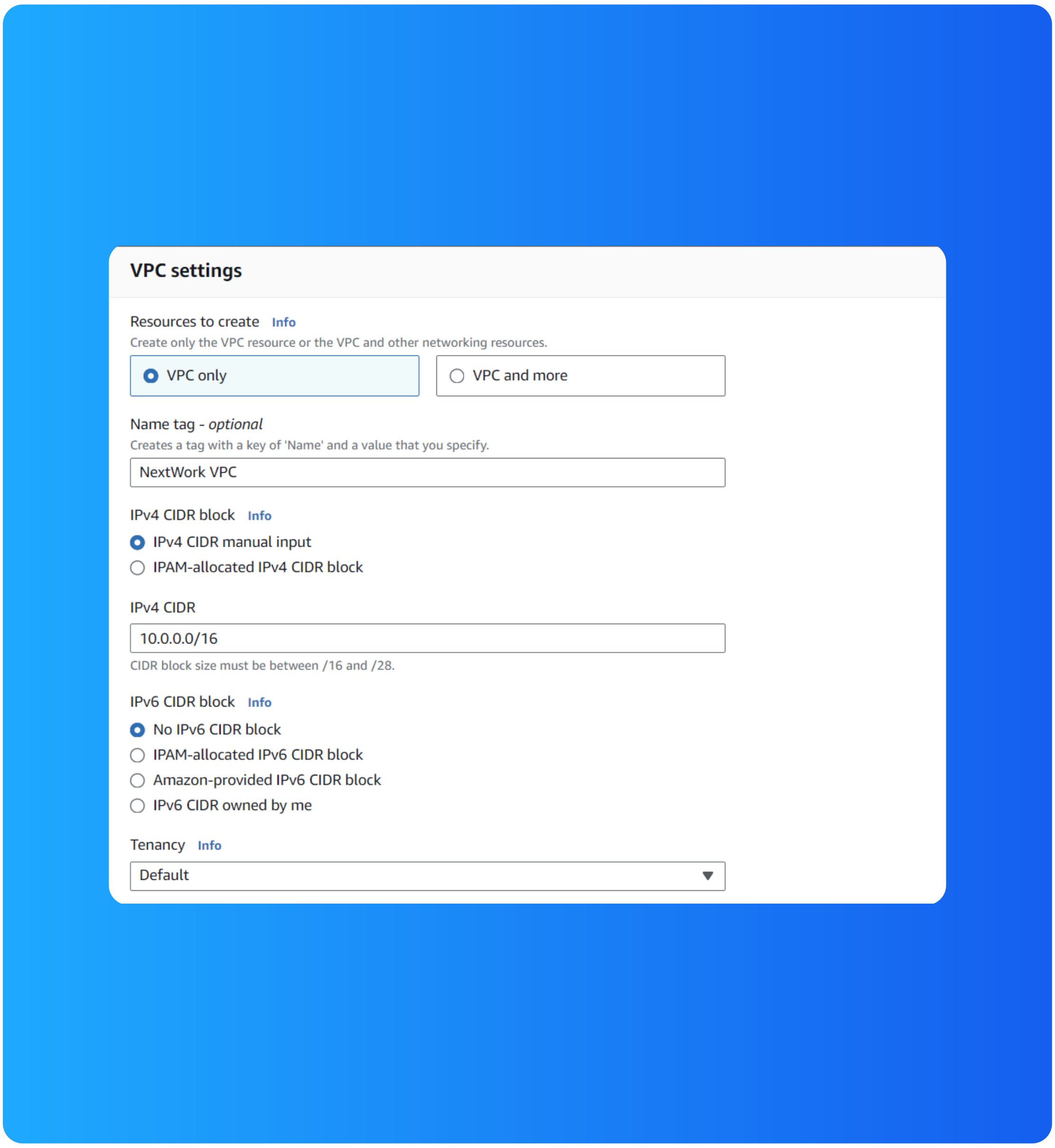

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network in AWS that allows you to isolate and control your cloud resources. It’s useful because it provides flexibility to define IP ranges, subnets, security, and routing, ensuring secure, scalable, and customizable network.

How I used Amazon VPC in this project

I used Amazon VPC in today’s project to create isolated subnets for public and private resources. I configured routing for secure internet access via an Internet Gateway, applied custom security groups, and ensured proper traffic flow between instanc

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the complexity of configuring security groups. Balancing access permissions while ensuring security required careful planning, which highlighted the importance of network security in cloud environments.

This project took me...

It took me 40 minutes to complete this project.

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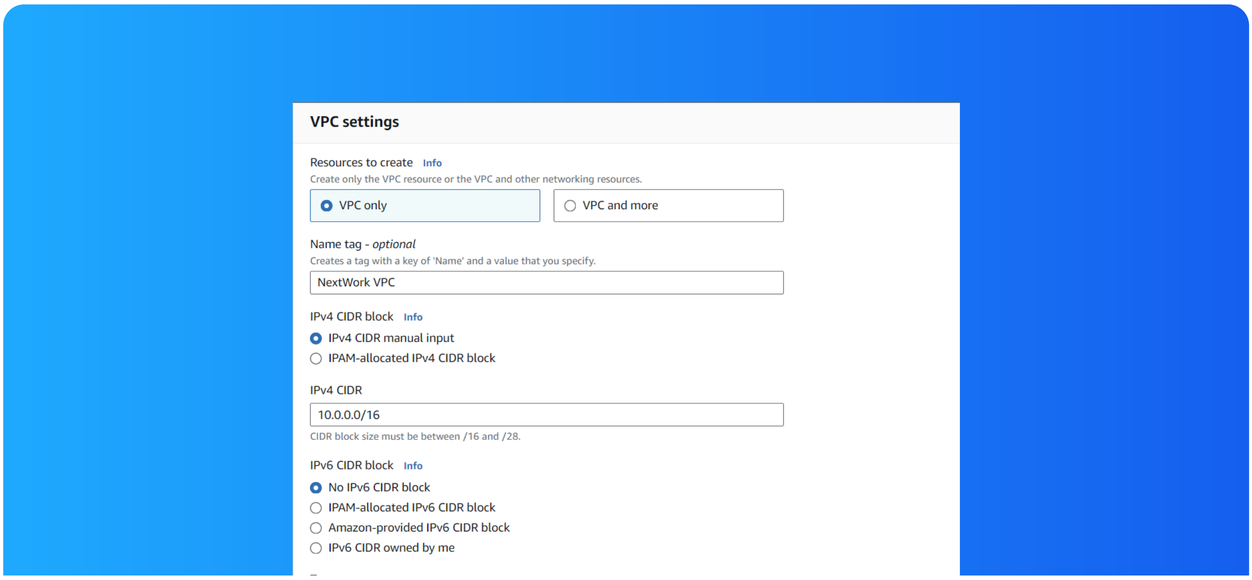
Virtual Private Clouds (VPCs)

VPCs are Virtual Private Clouds in AWS, providing isolated network environments where users can launch and manage AWS resources securely. Each VPC can define IP ranges, subnets, routing, and gateways, offering control over traffic flow and network.

There was already a default VPC in my account ever since my AWS account was created. This is because AWS automatically provides a default VPC for each region to simplify the process of launching resources, allowing basic networking features.

To set up my VPC, I had to define an IPv4 CIDR, which means specifying a range of IP addresses using CIDR notation (e.g., 192.168.0.0/16). This defines the network size and allows efficient allocation of IP addresses within my VPC for resources.



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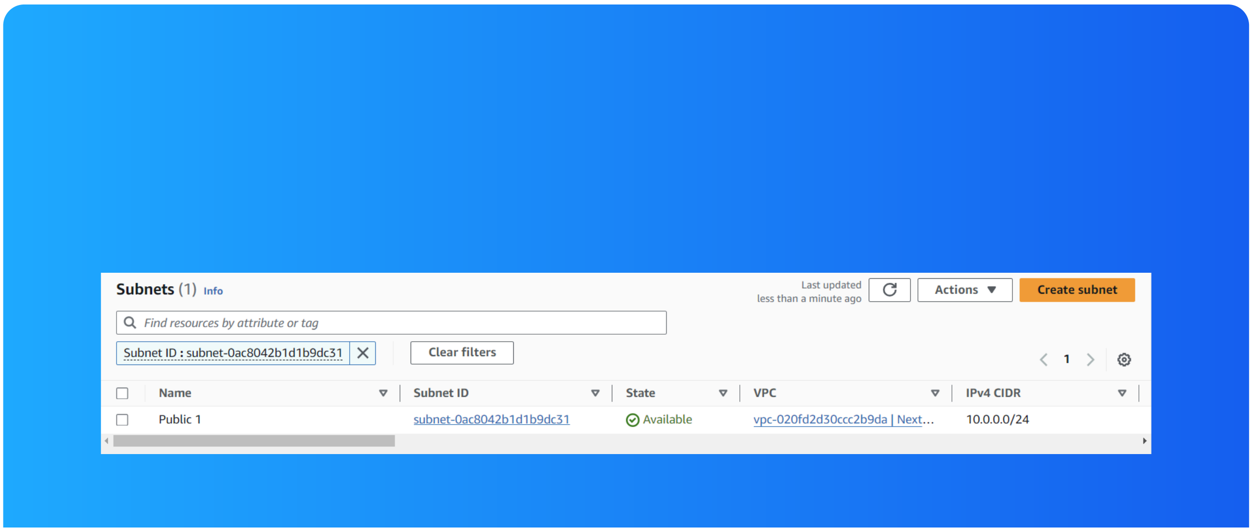
Subnets

Subnets are subdivisions of a VPC's IP address range that isolate resources within specific AZ. Each subnet allows control over traffic routing, security, and access, enabling more efficient network segmentation for public or private resource deploy.

There are already subnets existing in my account, one for every AZ in each region. AWS provides these default subnets to allow users to easily deploy resources across different zones with basic networking setup for high availability and redundancy.

I named my subnet Public 1, but that doesn’t automatically make my subnet a public subnet. For a subnet to be considered public, it has to be associated with a route table that directs traffic through an Internet Gateway, allowing external access.



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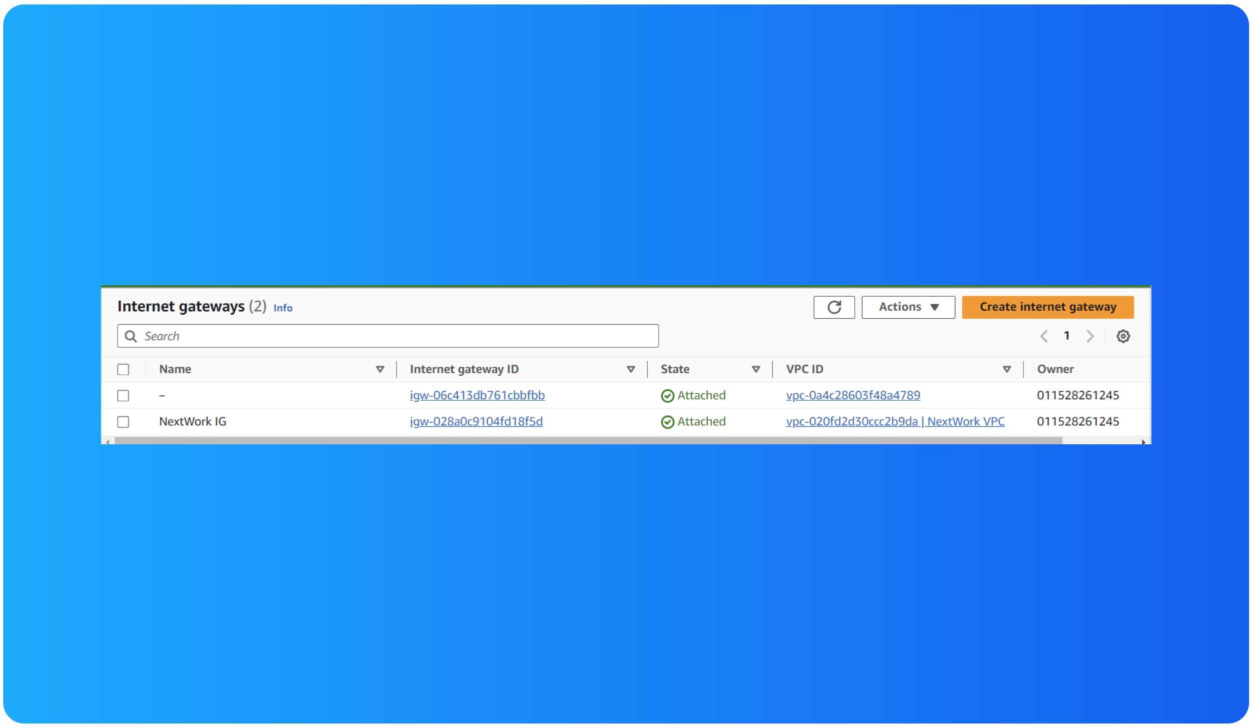
Internet gateways

Internet gateways are AWS components that allow communication between resources in a VPC and the internet. They provide a path for inbound and outbound traffic, enabling public access to instances within subnets configured with public IP addresses.

Attaching an internet gateway to a VPC means enabling the VPC to communicate with the internet. This allows resources, such as instances in public subnets, to send and receive traffic from external networks, making them accessible from outside VPC.



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